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VLSI implementation of a new LMS-based algorithm for noise removal in ECG signal

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ABSTRACT

Least mean square (LMS)-based adaptive filters are widely deployed for removing artefacts in electrocardiogram (ECG) due to less number of computations. But they possess high mean square error (MSE) under noisy environment. The transform domain variable step-size LMS algorithm reduces the MSE at the cost of computational complexity. In this paper, a variable step-size delayed LMS adaptive filter is used to remove the artefacts from the ECG signal for improved feature extraction. The dedicated digital Signal processors provide fast processing, but they are not flexible. By using field programmable gate arrays, the pipelined architectures can be used to enhance the system performance. The pipelined architecture can enhance the operation efficiency of the adaptive filter and save the power consumption. This technique provides high signal-to-noise ratio and low MSE with reduced computational complexity; hence, it is a useful method for monitoring patients with heart-related problem.

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Electrocardiogram (ECG); variable step-size delayed least mean square (VSS-DLMS) algorithm; mean square error (MSE); field programmable gate array (FPGA)

1. Introduction

Electrocardiogram (ECG) signals are generated by the electrical activity of heart muscles and provide an effective way to identify heart problems in biomedical applications. In general, ECG signals are measured by surface electrodes that are placed on different locations of the body. They are often corrupted with various types of noise, including baseline drift, motion artefacts, muscle contraction and power-line interference, etc. (Poungponsri & Yu, 2013). Presence of these noises in ECG may mislead the physician and result in wrong diagnosis. Hence, it is vital to eliminate these noises for an accurate diagnosis. However, the time-varying and non-stationary nature of the ECG noise makes the noise removal a complex task. Low-power, miniaturised and low-cost monitoring devices are the key components in portable ECG devices. The performance of these devices depends on analog signal conditioning, which is necessary to extract and amplify extremely low-level signals from a noisy environment (Joshi, Thaker, & Baghini, 2011).

The frequency components of the QRS complex are normally interfered by the high-frequency signals. Therefore, the interpretation of ECG signal parameters will be a tough task. Since ECG signals are non-stationary in nature, a filter has to be designed that changes its weights or coefficients according to the requirement. Several filtering techniques have been applied for ECG signal analysis, which includes adaptive and non-adaptive techniques (Laguna et al., 1996; Rahman, Shaik, & Reddy, 2011). In recent years, adaptive filters have become the key functional block in the field of digital signal processing (DSP) and digital communication (Ahmad, Kukrer, & Hocanin,

2011). Even though many adaptive algorithms have been proposed in the literature, the least mean square (LMS) algorithm is widely used due to its simplicity.

The LMS algorithm is ineffective in real-time applications due to its poor tracking ability (Lin & Wu, 1996). The recursive least square algorithm converges faster, but its computational complexity is higher than LMS-based algorithms (Iglesias, 2011). To improve the tracking capability and better convergence, a normalised step size has been introduced in the LMS algorithm, which is known as normalised LMS (NLMS) algorithm. Since the adaptive filtering performance depends on the step size, variable step-size algorithms have been proposed (Gupta & Joshi, 2008; Ni & Li, 2010).

The high computational requirement has limited the application of the adaptive algorithms in the past. However, with the advances in very large-scale integration (VLSI) technology, it is possible to implement complex circuits on a single chip (Rahimunnisa, Karthigaikumar, Kirubavathy, Jayakumar, & Kumar, 2013). Li and Mathews (1999) presented a computationally efficient block-adaptive algorithms suited for VLSI implementation. It has been proved that the delay in the coefficient adaptation path does not significantly affect the steady state behaviour (Long, Ling, & Proakis, 1989). Hence, the concept of delayed LMS (DLMS) algorithm has been utilised in this paper for computational complexity reduction with reasonable steady-state behaviour. Field programmable gate arrays (FPGAs) are being increasingly used for a wide variety of computing applications, especially in the area of DSP (Iglesias, 2011; Passas, Freear, & Fawcett, 2010). Due to rapid advancements in fabrication technology, modern FPGAs contain a large number of configurable logic blocks (CLBs) to perform a wide range of filtering applications. In this work, a variable step-size delayed LMS (VSS-DLMS) algorithm with pipelined architecture is proposed for adaptive FIR filters. FPGA implementation of pipelined VSS-DLMS filter achieves low-latency and low-power design.

This paper is organised as follows. Section 2 explains the different types of LMS algorithms. Section 3 explains the proposed work with necessary block diagram. Section 4 analyses the results obtained using the proposed method and compares them with those obtained by the existing techniques. Finally, some concluding remarks are presented in Section 5.

2. LMS algorithms

The LMS algorithm is one of the most widely used algorithms in adaptive filtering. The simplicity of the LMS algorithm has made it an important benchmark for other algorithms (Haweel, Gharieb, & Ibrahim, 1995). Figure 1 shows the adaptive filter structure. In this simple adaptive filter structures, LMS algorithm is applied to update filter coefficients. It utilises the gradient vector of the filter tap weights to converge on the optimal wiener solution.

On each of the iteration, the filter coefficients of the adaptive filter are updated according to the equation

$$w(n+1) = w(n) + \mu x(n)e(n), \quad (1)$$

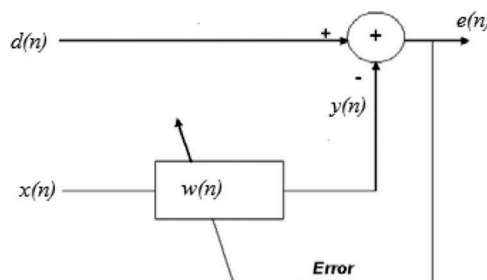


Figure 1. Block diagram of an adaptive FIR filter.

where $w(n+1)$ is the updated weight, $w(n)$ denotes old weight, $x(n)$ is the input, $e(n)$ is the error and μ is the step size that controls the convergence speed and stability. The μ value plays a significant role in judging the filtering performance of the adaptive filter. If the μ value is too small, the adaptive filter takes more time for converging to the optimal solution. On the other hand, if the μ value is too large, the stability of adaptive filter will be affected and its output diverges. The filter output $y(n)$ is given by

$$y(n) = w^t(n) \times (n), \quad (2)$$

where $w^t(n)$ indicates the weights in transposed form. The difference between the desired response $d(n)$ and obtained response $y(n)$ is given by

$$e(n) = d(n) - w^t(n) \times (n). \quad (3)$$

To analyse the noise removal capability of the filter, the input signal $s_1(n)$ is assumed to be corrupted with the noise signal $p_1(n)$:

$$E[e^2(n)] = E\{[s_1(n) - y(n)]^2\} + E[p_1^2(n)]. \quad (4)$$

The best least-squares estimate of the signal $s_1(n)$ can be obtained by reducing the mean square error (MSE). The adaptive filter eliminates the noise by iteratively lessening the MSE between the primary and the reference inputs. The computation time of the filter mainly depends on the multiply and accumulate (MAC) operations. The polarity of the error signal may be utilised to reduce the MAC operations.

The update equation of signed LMS algorithm is given by

$$w(n+1) = w(n) + \mu x(n) \text{sgn}\{e(n)\}. \quad (5)$$

The main drawback of the LMS algorithm is the utilisation of fixed step-size parameter. The NLMS is an improved version of the LMS algorithm that has normalised step-size value for each iteration of the algorithm. The chosen step size is proportional to the inverse of the total expected energy of the coefficients of the input signal vector $x(n)$ (Chinaboina et al., 2011; Sundeep & Kumari, 2012). The iterative procedure is very much similar to that of the LMS algorithm, which is described as follows:

Step 1: The output of the adaptive filter is given by the equation

$$y(n) = \sum_{i=0}^{N-1} w(n)x(n-i) = w^t(n)x(n). \quad (6)$$

Step 2: Error signal is computed by the difference between the desired output and the filter output.

$$e(n) = d(n) - y(n). \quad (7)$$

Step 3: The weight update equation of NLMS algorithm is given by

$$w(n+1) = w(n) + \left(\frac{\mu}{\delta + x^t(n)x(n)} \right) x(n)e(n), \quad (8)$$

where δ is a small positive constant to avoid large step-size parameter.

Each of the iterations of the NLMS algorithm requires $3N+1$ multiplications, slightly more than the standard LMS algorithm (Chinaboina et al., 2011; Sundeep & Kumari, 2012). The NLMS algorithm has far greater stability with unknown signals; however, its computational complexity makes it difficult to implement in real-time processing (Douglas & Meng, 1994). Pipelined NLMS (PINLMS) algorithm can be used to reduce the computational complexity (Ting, Woods, & Cowan, 2005). In pipelined filters, any operation on a long critical path is broken into levels of smaller, quicker

operations, with registers between levels, so as to get a smaller critical path delay and higher throughput.

Each of the iterations of PIPNLMS algorithm involves the following steps.

Step 1: The output of the adaptive filter is expressed as

$$y(n) = x_1(n) - d(n). \quad (9)$$

Step 2: The desired signal is estimated as

$$\hat{d}_n = x(n) \cdot \hat{w}_n. \quad (10)$$

Step 3: Error signal is computed by the difference between the estimated signal and the filter output.

$$e(n) = \hat{d}_n - y(n). \quad (11)$$

Step 4: The filter tap weights are updated by

$$w(n) = w^T(n - m_2) + \sum_{i=0}^{m_2-1} \mu \cdot e(n - m_3 - i) \cdot u(n - m_1 - i) \quad (12)$$

$$en = x_2(n) \cdot w^T(n - m_2) - x_1(n) + d(n), \quad (13)$$

where m_1 , m_2 and m_3 are the number of delays in input, weight update loop and error feedback loop, respectively.

3. Proposed method

A VSS-DLMS algorithm with pipelined architecture is proposed here for noise removal in ECG signal. The performance of pipelined filters is restricted due to feedback loops, hence the expected performance may not be realised (Ting et al., 2005). For this reason, the delayed filter implementation got attention of the researchers (Long et al., 1989), as shown in Figure 2. In this algorithm, first the filter output $y(n)$ is calculated, then the difference between the desired signal $d(n)$ and filter output $y(n)$ is computed to calculate the error signal. This error $e(n)$ is used to update the filter coefficients before the next sample arrives. In pipelined filters, a long critical path is broken into levels of smaller, so as to get a smaller critical path delay and a higher throughput.

Figure 3 depicts the adaptive filter with VSS-DLMS algorithm, comprising of N-tap FIR filter, coefficient update algorithm and delay blocks. The delayed error signal $e(n - mD)$ is applied to the VSS-LMS algorithm block that has a variable step size $\mu(n)$.

The VSS-LMS algorithms, in contrast to the NLMS algorithms, utilise a small step size initially and the value can be increased according to the requirement (Akhtar, Abe, & Kawamata, 2006). The

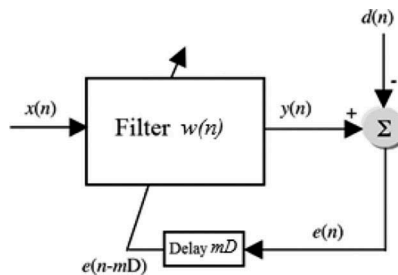


Figure 2. Implementation of delayed LMS algorithm.

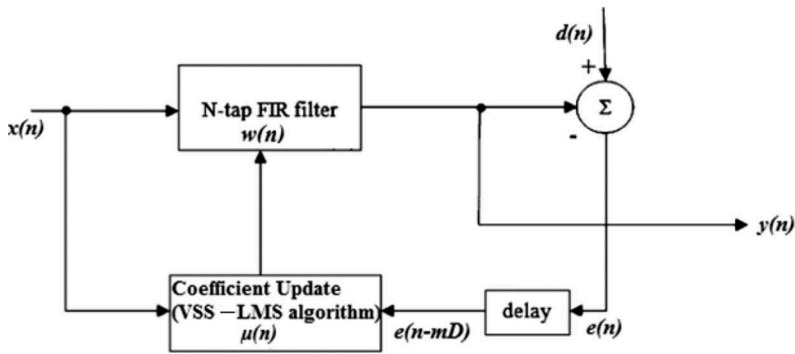


Figure 3. Block diagram of the proposed algorithm.

weight update in VSS-LMS algorithm described in the literature takes the form (Aboulnasr & Mayyas, 1999)

$$w(n+1) = w(n) + \mu(n)e(n)x(n), \quad (14)$$

where $\mu(n)$ is the variable step size applied to the filter.

It has a larger step size when the error is large and has a smaller step size when the error is small. The step size is computed in one of the following two ways (Gelfand, Wei, & Krogmeier, 1999): (1) a prior step size that depends on the past input data and errors and (2) a posterior step size that depends on the present input, past input data and errors. In addition, a bound is enforced on the step size to ensure stability of the filter.

The proposed algorithm utilises delay in the VSS-LMS algorithm. The DLMS algorithm can be described in three parts: weight update, error signal calculation and output signal estimation (Ting et al., 2005):

$$w(n+1) = w(n) + \mu e(n-mD)x(n-mD), \quad (15)$$

$$e(n-mD) = d(n-mD) - y(n-mD), \quad (16)$$

$$y(n-mD) = w^T(n-mD)x(n-mD), \quad (17)$$

where mD indicates the number of pipelining delays.

The condition to be satisfied for the DLMS algorithm to converge is described in Equation (18), in terms of the tap-input power (P_{TI}), filter order N and the pipelining stages m (Ting et al., 2005):

$$0 < \mu < \frac{2}{(N-2m-2)(P_{\text{TI}})}. \quad (18)$$

The critical path computation time of the system can be reduced by pipelining. Hence, the pipelining registers are inserted into the error feedback path to reduce the critical path of the long feedback loop of the recursive system (Ting et al., 2005).

The first moments of the weight error vector can be evaluated by

$$\varepsilon(n) = w(n) - \omega_0. \quad (19)$$

The vector ' ω_0 ' is the Wiener solution to which the algorithm should converge. Using VSS-LMS algorithm principles and delayed LMS equations (15)–(17), the VSS-DLMS equations are derived as follows.

The error estimate is calculated by

Table 1. Comparison in terms of computational complexity.

Algorithm	VSS-LMS	TDLMS	TDVSS	VSS-DLMS
Multiplications	$2N$	$6N + 1$	$6N + 4$	$2N$
Additions	$2N + 1$	$3N$	$3N + 2$	$2N + 1$
Delay	0	0	0	1

$$\bar{e}(n - D) = y(n - D) - w(n)x(n - D), \quad (20)$$

If the actual system output $y(n - D)$ is not available, the modified error can be computed as

$$\bar{e}(n - D) = e(n - D) + \hat{y}(n - D) - w(n)x(n - D), \quad (21)$$

where $\hat{y}(n - D)$ is the estimated output. From Equations (20) and (21), the modified update equation is written as

$$w(n + 1) = w(n) + \bar{e}(n - D)\mu(n)x(n - D). \quad (22)$$

The computational complexity of the proposed technique is compared with the VSS-LMS algorithm, transform domain LMS (TDLMS) algorithm and transform domain VSS-LMS (TDVSS) algorithm. The number of additions, multiplications and delay elements required for these algorithms is listed in Table 1 in terms of the filter order N . From the table, the proposed algorithm requires $2N$ multiplications, $2N + 1$ additions and one delay element that are far less than the transform domain algorithms.

4. Results and discussion

In this section, simulations are carried out to verify the functionality of the proposed method for ECG noise removal. The performance of the proposed VSS-DLMS algorithm is compared to the VSS-LMS, TDLMS and TDVSS algorithms. In these simulations, the step-size value of 0.1 is considered for the adaptive filter and the length of the filter is '5'. To evaluate the noise removal capability of the proposed filter, a random noise with variance of 0.04 has been added with the original ECG signals. The input signal-to-noise ratio for the non-stationary noise considered for the experiment is 1.25 dB. The number of iterations for the experiment is 5000. Ten sample data (record 101, record 103, record 104, record 107, record 117, record 119, record 122, record 209, record 219 and record 220) of the Massachusetts Institute of Technology-Beth Israel Hospital database have been used. Figure 4 shows the extraction of clear ECG form corrupted ECG data (record 106). However, the

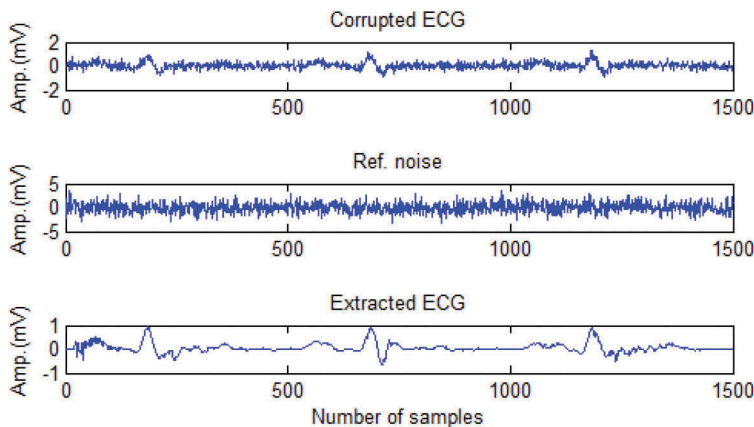


Figure 4. ECG noise removal using the VSS-DLMS algorithm.

obtained output is not enough to declare the efficiency of the VSS-DLMS algorithm. Hence, the MSE is calculated in accordance with the number of iterations.

To evaluate the performance of the proposed algorithm, VSS-LMS, TDLMS and TDVSS filters are compared with the designed filter in terms of convergence speed and steady-state error. The comparison in Figure 5 shows the convergence performance of VSS-DLMS is slightly better than other algorithms for noisy inputs in terms of the steady-state error and convergence speed. Among the existing algorithms, TDVSS gives the MSE of 12.8 in 500 iterations, while VSS-DLMS provides the MSE of 17.24 in 500 iterations which is the performance improvement of 26.2%. The proposed algorithm can be implemented using pipelined architectures to enhance the operation efficiency of the adaptive filter and also save the power consumption.

The hardware description language (HDL) simulation is vital for implementing the adaptive filter in FPGAs and application specific integrated circuits (ASICs). ASIC chips are fully customised chips which are designed according to the user specifications. The speed, area and power can be optimised in these chips. Conversely, FPGAs are based on semicustom technology with standard CLBs. Nowadays, FPGAs are used for a wide variety of computational applications, especially in the area of DSP (Ting et al., 2005). The drawbacks of FPGAs are limited number of logic operations, speed limitation and less number of signal routing options for connecting logical operators on the array. Two popular HDLs are Verilog HDL and very high speed integrated circuit hardware description language. The Verilog HDL is popular among designers because of the availability of many integrated circuit-related functions, data types and large number of built-in libraries. Adaptive filter using the proposed algorithm with pipelined architecture is considered. Modelsim tool is used for simulation. The coefficients or weights of the adaptive FIR filter can be updated using the VSS-DLMS algorithm. The output of the filter is compared with desired output and an error signal is calculated. The coefficients are updated by adding the current coefficients with learning rate parameter. The updated coefficients are differed based on the learning rate parameter. The simulation of the HDL implementation with filter coefficients and the step size is shown in Figure 6.

The next phase in the ASIC design flow is converting RTL code to the gate-level net list, called synthesis, targeted to a specific technology. Synthesis is a process in which the RTL code is transformed to the gate-level net list. Optimisation constraints include operating frequency (clock period), input and output delays at the I/Os. Operating temperature, process variations, supply voltage and wire load models come under environmental constraints (Murgai, Brayton, & Sangiovanni-Vincentelli, 1995). TSMC 65 nm technology is selected as a target library, and

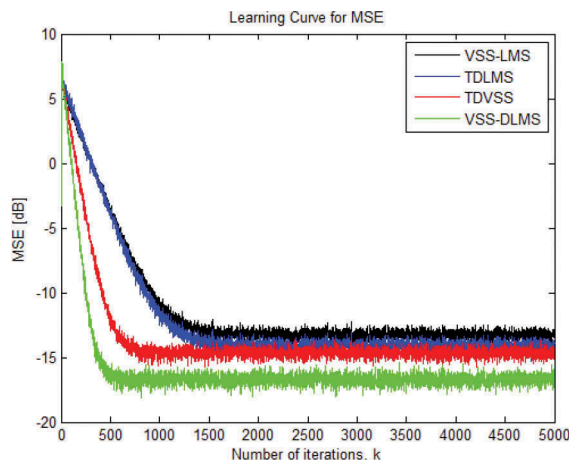


Figure 5. Comparison of MSE for input white noise.

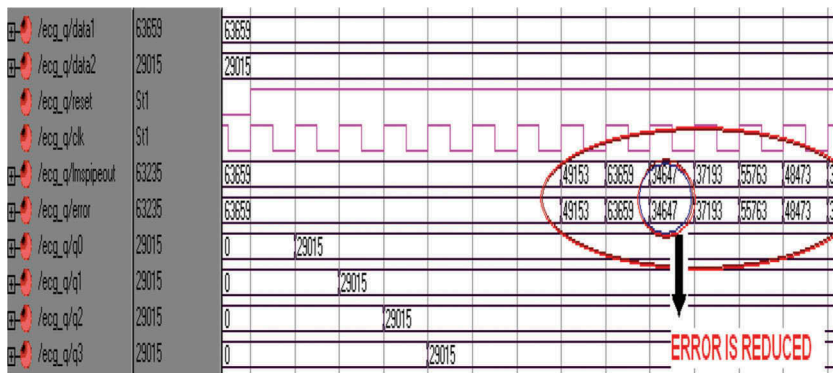


Figure 6. Simulation of filter coefficients and step size.

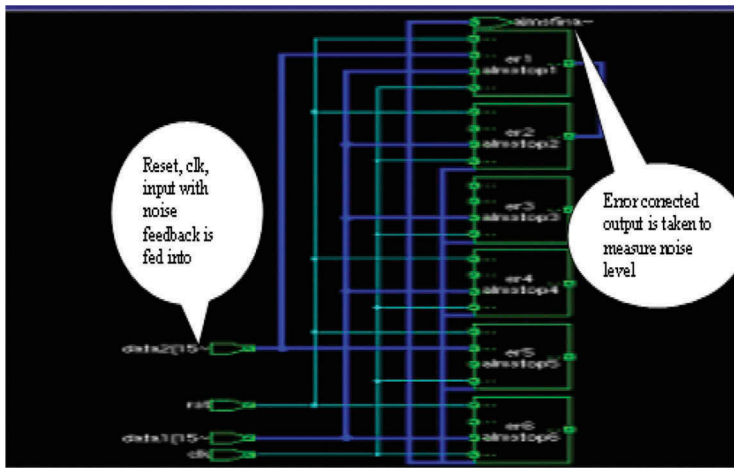


Figure 7. Synthesised schematic of the VSS-DLMS filter.

Synopsys Design Compiler tool is used for synthesis. Figure 7 shows the schematic, which was generated after synthesising the RTL code. The final stage in the design process is routing all the cells in the core area and to the I/O cells. Routing is a two-step process, first global routing is carried out and then detailed routing is performed. This ensures that all the cells are interconnected as per the net list obtained during synthesis. And, at the same time, the timing is also met. Figure 8 shows the final chip of the proposed filter.

The results obtained for the proposed VSS-DLMS filter with pipelining and low-power concepts were incorporated to outperform their counterpart without pipelining. Table 2 provides the comparison of the proposed filter without pipelining and with pipelining. The target device for the proposed filter is Xilinx XC5VLX110-1 FF676. The architecture without pipeline requires less cell area and consumes 182.96 mW of power at 1 pF capacitance load. The pipelined filter reduces the dynamic power by 13.14% and cell leakage power by 23.51% at the expense of 7.37% increase in total die size. However, the extra die area could be reduced to a large extent by adopting optimised ASIC design methodology with proper floor planning, placement and routing.

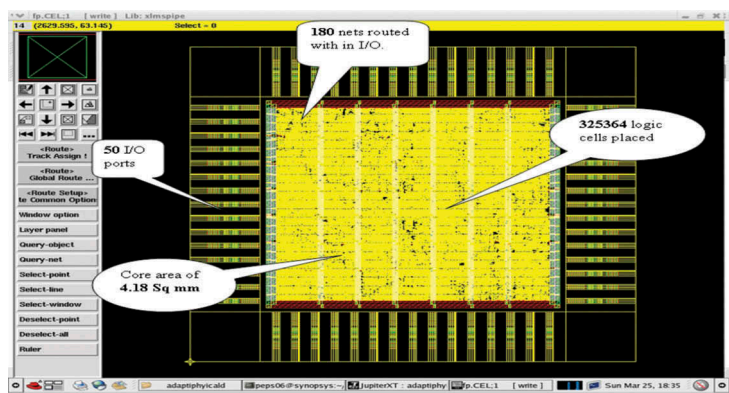


Figure 8. Final chip of the VSS-DLMS filter.

Table 2. Chip report comparison.

S. no.	Parameters	Without pipelining	With pipelining
1	Maximum critical paths	128	521
2	Number of ports	50	50
3	Number of nets	232	180
4	Total area sq. micron	264,869.42	284,401.32
5	Logic cells placed	312,435	325,364
6	Dynamic power	182.96 mW	160.77 mW
7	Cell leakage power	495.48 μ W	383.96 μ W

5. Conclusion

We proposed a VSS-DLMS adaptive filter to remove the noise from ECG signal for improved feature extraction in ECG signal. The proposed algorithm requires $2N$ multiplications, $2N + 1$ additions and one delay element that are far less than the transform domain algorithms. The dynamic power and cell leakage power are reduced by 13.14% and 23.51% respectively with the slight increase in the total die size while implementing in a Xilinx XC5VLX110-1 FF676 device.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

Aboulnasr, T., & Mayyas, K. (1999). Complexity reduction of the NLMS algorithm via selective coefficient update. *IEEE Transactions on Signal Processing*, 47(5), 1421–1424. doi:10.1109/78.757235

Ahmad, M., Kukrer, O., & Hocanin, A. (2011). Recursive inverse adaptive filtering algorithm. *Digital Signal Processing*, 21(4), 491–496.

Akhtar, M. T., Abe, M., & Kawamata, M. (2006). A new variable step size LMS algorithm-based method for improved online secondary path modeling in active noise control systems. *IEEE Transactions on Audio, Speech and Language Processing*, 14(2), 720–726. doi:10.1109/TSA.2005.855829

Chinaboina, R., Ramkiran, D. S., Khan, H., Usha, M., Madhav, B. T. P., Srinivas, K. P., & Ganesh, G. V. (2011). Adaptive algorithms for acoustic echo cancellation in speech processing. *International Journal of Research & Reviews in Applied Sciences*, 7(1), 38–42.

Douglas, S. C., & Meng, T. H.-Y. (1994). Normalized data nonlinearities for LMS adaptation. *IEEE Transactions on Signal Processing*, 42, 1352–1365. doi:10.1109/78.286952

Gelfand, S. B., Wei, Y., & Krogmeier, J. V. (1999). The stability of variable step-size LMS algorithms. *IEEE Transactions on Signal Processing*, 47(12), 3277–3288. doi:10.1109/78.806072

- Gupta, A., & Joshi, S. (2008). Variable step-size LMS algorithm for fractal signals. *IEEE Transactions on Signal Processing*, 56(4), 1411–1420. doi:[10.1109/TSP.2007.909374](https://doi.org/10.1109/TSP.2007.909374)
- Haweel, T. I., Gharieb, R. R., & Ibrahim, H. M. (1995). Median least mean squares lattice (MLMSL) adaptive filters. *International Journal of Electronics*, 78(1), 101–111. doi:[10.1080/00207219508926142](https://doi.org/10.1080/00207219508926142)
- Iglesias, M. E. (2011). Implementation of QRD-RLS algorithm on FPGA. Application to noise canceller system. *IEEE Latin America Transactions*, 9(4), 458–462. doi:[10.1109/TLA.2011.5993728](https://doi.org/10.1109/TLA.2011.5993728)
- Joshi, S., Thaker, V., & Baghini, M. S. (2011). Versatile ultra low noise low power analog signal conditioning chip with integrated drivers. In *2011 13th international symposium on Integrated circuits (ISIC)* (pp. 114–117). Singapore: IEEE.
- Laguna, P., Jané, R., Olmos, S., Thakor, N. V., Rix, H., & Caminal, P. (1996). Adaptive estimation of QRS complex wave features of ECG signal by the Hermite model. *Medical & Biological Engineering & Computing*, 34(1), 58–68. doi:[10.1007/BF02637023](https://doi.org/10.1007/BF02637023)
- Li, L., & Mathews, V. (1999). Efficient block-adaptive parallel-cascade quadratic filters. *IEEE Transactions on Circuits and Systems II: Analog and Digital Signal Processing*, 46(4), 468–472. doi:[10.1109/82.755419](https://doi.org/10.1109/82.755419)
- Lin, S. S., & Wu, W. R. (1996). The ASIC design of an LMS-based decision feedback equalizer for TDMA digital cellular radio. In *Seventh IEEE international symposium on Personal, Indoor and Mobile Radio Communications, 1996. PIMRC'96* (Vol. 1, pp. 218–222). Taipei: IEEE.
- Long, G., Ling, F., & Proakis, J. (1989). The LMS algorithm with delayed coefficient adaptation. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 37(9), 1397–1405. doi:[10.1109/29.31293](https://doi.org/10.1109/29.31293)
- Murgai, R., Brayton, R. K., & Sangiovanni-Vincentelli, A. (1995). *Logic synthesis for field-programmable gate arrays* (Vol. 324). Berlin: Springer Science & Business Media.
- Ni, J., & Li, F. (2010). A variable step-size matrix normalized subband adaptive filter. *IEEE Transactions on Audio, Speech, and Language Processing*, 18(6), 1290–1299. doi:[10.1109/TASL.2009.2032948](https://doi.org/10.1109/TASL.2009.2032948)
- Passas, G., Freear, S., & Fawcett, D. (2010). The effect of structural design parameters on FPGA-based feed-forward space-time trellis coding-orthogonal frequency division multiplexing channel encoders. *International Journal of Electronics*, 97(8), 907–926. doi:[10.1080/00207211003733346](https://doi.org/10.1080/00207211003733346)
- Poungponsri, S., & Yu, X.-H. (2013). An adaptive filtering approach for electrocardiogram (ECG) signal noise reduction using neural networks. *Neurocomputing*, 117, 206–213. doi:[10.1016/j.neucom.2013.02.010](https://doi.org/10.1016/j.neucom.2013.02.010)
- Rahimunnisa, K., Karthigaikumar, P., Kirubavathy, J., Jayakumar, J., & Kumar, S. S. (2013). A 0.13- μm implementation of 5 Gb/s and 3-mW folded parallel architecture for AES algorithm. *International Journal of Electronics*, 101(2), 1–12.
- Rahman, M. Z. U., Shaik, R. A., & Reddy, D. R. K. (2011). Efficient sign based normalized adaptive filtering techniques for cancelation of artifacts in ECG signals: Application to wireless biotelemetry. *Signal Processing*, 91(2), 225–239. doi:[10.1016/j.sigpro.2010.07.002](https://doi.org/10.1016/j.sigpro.2010.07.002)
- Sundeeep, G., & Kumari, U. R. (2012). Reduction of power line interference by using adaptive filtering techniques in electrocardiogram. *International Journal of Innovative Technology and Exploring Engineering*, 1(5), 83–86.
- Ting, L., Woods, R., & Cowan, C. (2005). Virtex FPGA implementation of a pipelined adaptive LMS predictor for electronic support measures receivers. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 13(1), 86–95. doi:[10.1109/TVLSI.2004.840403](https://doi.org/10.1109/TVLSI.2004.840403)